

Crestline Business Online - Wire Center: Current-State Architecture

Existing commercial portal wire capabilities, integrations, status model and constraints

Document ID	CBO-ARCH-WC-4.1
Version / Status	4.1 / Approved
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Effective / Last reviewed	Last reviewed 2026-08-20
Classification	INTERNAL - CONFIDENTIAL
Related documents	PPH-API-CAT-2026.3; ENS-INT-3.2; ADR-PAY-019; PIR-2024-07; SEC-STD-22

1. Overview

Crestline Business Online (CBO) is CNB's commercial digital banking portal (web and mobile) serving Treasury Management clients. The **Wire Center** module provides domestic (Fedwire) and international (Swift) wire initiation, templates, dual approval, release with step-up authentication, a wire activity list, wire detail, CSV export and confirmation PDF download. Wire Center does not currently provide milestone tracking, international (gpi) status, hold explanations or client actions on held wires.

1.1 Usage (trailing 90 days to 2026-07-31)

Metric	Value
Client companies with Wire Center entitlement	9,800
Active users (90-day)	41,000
Wires initiated via CBO per business day	11,600 avg (9,900 domestic / 1,700 international); month-end peak ~19,000
Wire Center page views per month	1.4 million
'Refresh status' clicks per month	310,000
Share of CNB outgoing wires originated in CBO	~67% (remainder: host-to-host files, Swift for Corporates, branch, Ops)

2. Logical architecture

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[Wire Center MFE (React 18, Aurora DS v4)]
  | HTTPS (OAuth2 session)
[cbo-wire-bff (Java 21 / Spring Boot 3, OpenShift)] ---> [CES /ces/v2/users/{id}/entitlements]
  | REST (sync) | REST (sync) \---> [CBO Auth /cbo/auth/v2/step-up]
[PPH v1 APIs] [ENS /ens/v3/notifications]
  /pph/v1/wires/{ref}/status (EP-PPH-01)
  /pph/v1/wires?clientId... (EP-PPH-02)
  POST /pph/v1/wires (EP-PPH-03)
```

Not used by Wire Center today: CBO Status Projection Service (SPS, ACH only), pay.wire.lifecycle.v2, PPH v2 APIs, gpi data, HMS APIs, TDIP Insights API

2.1 Integration inventory

Consumer	Endpoint / interface	Purpose	Version	Notes
cbo-wire-bff	POST /pph/v1/wires (EP-PPH-03)	Submit approved wire	v1	Returns pphId; stored with cboRef
cbo-wire-bff	GET /pph/v1/wires?clientId... (EP-PPH-02)	Activity list	v1	Max 500 rows; filters applied client-side (CBO-4302)
cbo-wire-bff	GET /pph/v1/wires/{ref}/status (EP-PPH-01)	Wire detail status	v1	Cached 60 s; refresh throttled 1/60 s per wire
cbo-wire-bff	GET /ces/v2/users/{id}/entitlements (EP-CES-01)	Authorization	v2	Account-level filtering of results
cbo-wire-bff	POST /cbo/auth/v2/step-up (EP-AUTH-01)	Step-up on release	v2	Push or OTP; reusable for other high-risk actions

Consumer	Endpoint / interface	Purpose	Version	Notes
cbo-wire-bff	POST /ens/v3/notifications (EP-ENS-01)	Wire notifications	v3	Produces TRS.WIRE.* events (Section 5)

3. Status model and client-facing labels

Wire Center derives the displayed status from PPH v1 status values. Statuses prior to submission (Draft, Pending Approval, Approved) are CBO-owned. The label '**Completed**' replaced 'Processed' in R26.1 (CBO-4402) following client feedback that 'Processed' was unclear.

PPH v1 status	CBO label	Visual	Since
(CBO) DRAFT / PENDING_APPROVAL / APPROVED	Draft / Pending Approval / Approved	grey	R21
RECEIVED	Submitted	blue	R21
PENDING	In Process	blue	R21
HELD	Pending Review	amber	R22
PROCESSED	Completed	green check	R26.1 (was 'Processed')
REJECTED	Rejected	red	R21
CANCELLED	Cancelled	grey	R21
RETURNED	Returned	red	R23

Hold reasons are not displayed today; held wires show 'Pending Review' with the standard text 'This wire is being reviewed.'

4. Identifiers stored by Wire Center

Identifier	Example	Source	Stored in CBO?
cboRef	CBW-20260924-004812	CBO	Yes (primary key)
pphId	PPH26092400481233	PPH v1 POST response	Yes
IMAD (Fedwire input reference)	20260924QMGFT015000123	PPH v1 fedRef	Displayed only (CBO-4355); not persisted
OMAD (Fed output reference)	-	Not available in v1	No
UETR (end-to-end tracking id)	-	Not available in v1	No

5. Notifications produced by Wire Center

Event type	Trigger	Template	Channels
TRS.WIRE.APPROVAL_REQUIRED	Wire awaiting approver	TPL-WIRE-APR-01	email, push, in-app
TRS.WIRE.RELEASED	PPH v1 status transitions to PROCESSED	TPL-WIRE-REL-02	email, push, in-app, SMS (opt-in)
TRS.WIRE.REJECTED	PPH v1 status REJECTED	TPL-WIRE-REJ-01	email, push, in-app

6. Security and entitlements

- Entitlement types in use: WIRE_VIEW, WIRE_INITIATE, WIRE_APPROVE, WIRE_RELEASE, WIRE_TEMPLATE_ADMIN (CES).
- Release requires step-up authentication (EP-AUTH-01) and client-configured dual approval above client thresholds.
- External sharing of client transaction data (links or documents shared with non-users) is not supported; any such capability requires an InfoSec design review under SEC-STD-22 (Digital Channel External Sharing & Step-Up Standard) and a Privacy Impact Assessment.
- Search results are filtered to the user's entitled accounts in the BFF.

7. Constraints and known limitations

- No status polling.** Per ADR-PAY-019 (following INC-2024-1182), Wire Center must not poll PPH for status; only user-initiated refresh is permitted (throttled 1 per 60 s per wire). New status capabilities must be event-driven.

2. **PPH v1 sunset 2027-03-31.** Migration to v2 is tracked as CBO-4471 (34 pts, unscheduled) and depends on CBO-4473 (CES account-filter mapping).
3. **No international tracking.** gpi status is visible only to Payment Operations in the Investigations Workbench (spike CBO-4480).
4. **Activity list limits.** v1 list returns max 500 rows without cursor; no beneficiary-name search server-side.
5. **Status Projection Service (SPS, CBO-3815)** is configured for ACH only; wire support requires a new mapping module and a consumer ACL on pay.wire.lifecycle.v2.

8. Reusable assets

Asset	Origin	Reuse notes
<cbo-journey-timeline> component (Aurora DS v4)	CBO-3802 (ACH Payment Tracker)	Rail-agnostic milestone timeline incl. terminal/error styling
Status Projection Service (SPS)	CBO-3815	Kafka consumer -> Postgres read model -> /cbo/sps/v1; add payment types by config + mapping module
Step-up authentication	CBO Auth v2	Already used for wire release; reusable for other high-risk client actions
ENS producer library	cbo-commons 3.x	Templated sends with masking helpers